

EXPERIMENT – 7

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/n1"
        android:hint="First"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/n2"
        android:hint="Second"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <Button
            android:id="@+id/add"
            android:text="+"
            android:layout_weight="1"
            android:layout_width="0dp"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/sub"
            android:text="-"
            android:layout_weight="1"
            android:layout_width="0dp"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</LinearLayout>
```

```
<Button
    android:id="@+id/mul"
    android:text="×"
    android:layout_weight="1"
    android:layout_width="0dp"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/div"
    android:text="÷"
    android:layout_weight="1"
    android:layout_width="0dp"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

```
<Button
    android:id="@+id/clear"
    android:text="C"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<TextView
    android:id="@+id/result"
    android:text="Result"
    android:textSize="18sp"
    android:textStyle="bold"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

JAVA CODE:

```
package com.example.calculatorapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.*;

public class MainActivity extends AppCompatActivity {
    EditText n1,n2;
    TextView result;
    Button add,sub,mul,div,clear;

    protected void onCreate(Bundle b){
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        n1=findViewById(R.id.editTextNumber1);
        n2=findViewById(R.id.editTextNumber2);
        result=findViewById(R.id.textViewResult);

        add=findViewById(R.id.buttonAdd);
        sub=findViewById(R.id.buttonSubtract);
        mul=findViewById(R.id.buttonMultiply);
        div=findViewById(R.id.buttonDivide);
        clear=findViewById(R.id.buttonClear);

        add.setOnClickListener(v->calc('+'));
        sub.setOnClickListener(v->calc('-'));
        mul.setOnClickListener(v->calc('*'));
        div.setOnClickListener(v->calc('/'));

        clear.setOnClickListener(v->{
            n1.setText("");
            n2.setText("");
            result.setText("Result");
        });
    }
    void calc(char op){
        if(n1.getText().toString().isEmpty()||n2.getText().toString().isEmpty()){
            result.setText("Enter numbers");
            return;
        }
    }
}
```

```
double a=Double.parseDouble(n1.getText().toString());
double b=Double.parseDouble(n2.getText().toString());
double r=0;

switch(op){
    case '+': r=a+b; break;
    case '-': r=a-b; break;
    case '*': r=a*b; break;
    case '/': if(b==0){result.setText("Divide by 0"); return;} r=a/b;
}

result.setText("Result: "+r);
}
```

OUTPUT:

